



$$T_m \dot{\omega}_m(t) + \omega_m = K_m u_r(t)$$

$$\frac{\Omega_m(s)}{U_r(s)} = \frac{K_m}{T_m s + 1}$$

$$T_m \ddot{\theta}_m(t) + \dot{\theta}_m(t) = K_m u_r(t)$$

$$\frac{\Theta_m(s)}{U_r(s)} = \frac{K_m}{s(T_m s + 1)}$$